

"The economy is an emergent phenomenon, in which humans are the building blocks"

Traditional economics isn't working, but a radical alternative is now coming of age, as complexity theorist **J. Doyne Farmer** tells Thomas Lewton

IN 2006, economists at the Federal Reserve Bank of New York started to worry about the overheating US housing market. Concerned that the bubble might burst, they used their best model to predict what would happen if house prices dropped by 20 per cent. Not much, was the answer it churned out. Soon after, house prices fell by almost exactly this amount, leading to probably the worst period of global economic decline in a century.

Economics is often lambasted for being a pseudoscience, with dense mathematical formulae that belie its subjectivity and a poor track record of making accurate predictions. J. Doyne Farmer thinks we can do better. In his new book, *Making Sense of Chaos*, he unpicks why standard economic approaches often fail – and presents a radical alternative. Complexity economics, as it is called, treats economies as systems akin to natural ecosystems or Earth's climate. Giant computer simulations based on these ideas offer a better representation of how billions of people interact within the global economy.

Farmer currently holds posts at the University of Oxford and the Santa Fe Institute in New Mexico, but his journey into economics

has been unconventional. It began when he dropped out of graduate school, built the world's first wearable computer and used it to beat the casino at roulette. In the 1990s, he set up Prediction Company, where he applied similar principles to the stock market. A pioneer of chaos theory and complex systems, he believes that complexity economics has recently come into its own, making reliable predictions about the world's most intractable economic problems. It is time, he argues, for policy-makers to take note.

Thomas Lewton: Why do we need a more objective economics?

J. Doyne Farmer: One of the problems with mainstream economics now is that, on almost any important issue, there is a diversity of opinions about the right answer. Austerity is a good example. Following the financial crisis of 2008, Wolfgang Schäuble, Germany's finance minister, argued that austerity had to be imposed on the Greeks – that it was the only cure for their failing economy. Others, like Nobel-prizewinning economist Paul Krugman, argued that, no, you had to actually cut them some slack. When economists



disagree, they let politicians disagree. In the end, it means that we make our decisions based on gut feelings and beliefs rather than science. I think complexity economics offers the possibility for a more objective economics where the answers aren't begotten by the assumptions.

What are the problematic assumptions that traditional economics makes?

The core idea underlying standard economic theory is that people have utility functions, which score what they like and what they don't like. Often it is simply the fact that households like to consume – the more the better. Or businesses like to make profits – the more the better. These economic “agents” then make decisions that maximise their utility, meaning that they allow them to consume or make as much money as possible. But do we actually have the cognitive power needed to do the calculations that maximise our utility? In the traditional approach to economics, the answer is yes. This idea of *Homo economicus*, whereby people act completely rationally, dominated economics from roughly 1970 until about 2000. Then economists began to realise that *Homo economicus* is a poor model of what people do, at least in most circumstances.

Standard economics also assumes that economic systems, in the long term, settle down and reach equilibrium. For example, an economic market can determine the price for something by balancing supply and demand. But this is only a decent approximation of

Complexity economics can help us transition to sustainable energy



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reality in very simple cases. Economics is a bit stuck now because, on the one hand, it's clear that rationality is not the best model – nor is the idea of equilibrium, in most cases – but on the other hand, if you let go of rationality and equilibrium, the whole mathematical framework that was developed crumbles.

How does complexity economics differ in its approach?

We treat the economy as a complex system. This specifically means a system where there is emergent behaviour: the system as a whole exhibits properties that are not present in its building blocks themselves. An ant colony is an example of emergence. Each ant is a very simple creature, and yet ant colonies can farm mushrooms, ranch aphids and do all kinds of sophisticated behaviours. The economy is an emergent phenomenon, too, in which humans are the building blocks.

Complexity economics recognises that economies have quite a lot in common with other complex systems, such as ecosystems, biological evolution or even Earth's atmosphere. In complexity economics, we often borrow mathematical tools from other fields to build our models.

In what way is an economy like an ecosystem?

An ecosystem is an interaction of specialist agents: grass is a specialist at extracting energy from the sun and turning it into grass; zebras are specialists at turning grass into zebras; and lions are specialists at turning zebras into lions. Each of those entities strongly depends on the others. If you kill all the lions, that's very bad for the grass, because then the zebras eat all the grass. Similarly, in economies, we are



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specialised in the jobs we do and the kind of products we make. Predicting the economy is about understanding how all of these specialists interact with each other and what emerges out of the other end.

Practically, though, what do complexity economics models look like?

First of all, good models have what I call verisimilitude: they should follow the same basic principles that the real world follows. The models are agent-based and we try to understand how the agents actually make their decisions. You don't need to know everything about psychology, but it's nice to know as much as possible about how real people make economic decisions.

For example, you might simulate a market containing agents that buy and sell. They might make their decisions based on simple rules like “buy undervalued assets” or “look around at your neighbours to see what they're doing, and copy them if they're doing better than you”, or they might use trial and error. Then you can put these rules into a computer and simulate what happens over time. The agents input information and make decisions, their decisions change the economy, then the agents input new information. And you just keep going around and around that loop, letting the simulation evolve with time to see what it does.



"Economies have a lot in common with ecosystems or even Earth's atmosphere"

The ideas behind complexity economics have been around since the 1960s. What is driving the current advances in this field?

There are several factors. The most important one is computing power. Computers are now a billion times more powerful than in the 1960s. We also understand a lot more about how people behave and how they make their decisions. And we have vastly more data – which is partly a consequence of increased computer power.

In the past, our agent-based models were typically qualitative. Most of them would simulate an economy in a hypothetical world that had some realistic properties. Now, we're beginning to make models where we can make sharp quantitative predictions. For instance, we can say, for a specific economy, that next quarter's GDP will be a specific number. That's important because it means you can measure how well the model is performing and keep adjusting it to try to make it perform better. This makes the models more powerful. It allows us to keep score of how well we are doing and to go head-to-head with mainstream models.

How does the predictive power of complexity economics compare with that of standard economics?

Even though complexity economics is relatively new and has consumed relatively little human capital so far, there are already examples where

it is doing very well. I begin my book with the story of how, shortly after covid-19 struck, I assembled a research team to predict what the economic consequences of the pandemic would be for the UK. Using an agent-based model, we predicted that in the second quarter of 2020, the UK would take a GDP hit of 21.5 per cent. We made predictions before the answer was known. When the dust settled, the answer was 22.1 per cent. For comparison, the Bank of England's estimate was 30 per cent. We also calculated levels of unemployment and several other economic indicators pretty accurately – including what would happen to 52 different industry sectors. This provides a proof of principle that this can really work.

Is complexity economics already being used in the real world?

Some central banks have research teams that are beginning to develop agent-based models for the economy. Those include the Bank of England, the Bank of Canada, the Hungarian National Bank, the Bank of Spain and the Bank of Italy. These are smaller efforts within the larger research teams of those central banks, but it is something that's becoming established now.

What possible applications for complexity economics are you most excited about?

Climate change is a problem where complexity

Standard economics failed to predict the financial crash of 2008 (left and far left)

economics can have a big influence. We're going through transitions in our energy systems and food systems that are going to happen very quickly and are going to profoundly change the way we do things. While these are happening, we will be far from equilibrium, and standard economic approaches will be of limited value.

My colleagues and I are designing something called the Climate Policy Laboratory to try to model the different relevant parts of the economy. The model comprises modules that describe energy, agriculture, innovation, finance, production and labour. These can be used separately or together to see how changes in one affect the others. In this way, we will be able to test different policies and see how the transition would play out – for example, how workers will transition to different occupations, or what happens when, say, BP decides to divest from oil and invest in solar farms. The answers are not obvious because of all the indirect network effects. The goal is to make a more just transition as rapidly as possible.

Some economists see economic growth as part of the problem when it comes to climate change. What's your view?

I'm not a fan of no-growth economics. We are going to need a lot of resources to make the transition quickly. We have the technologies, and if we wanted to go on a wartime footing and were willing to make some sacrifices, we could do it all in three years. But people don't seem willing to do that. Pushing the no-growth agenda too hard could alienate people from wanting to take action about climate change and create even more problems.

I think there's a new narrative that's more productive, which is that climate change is a great business opportunity rather than a hot potato to be passed around at COP meetings. People are beginning to realise that whoever makes this happen will get rich. If we can harness this profit motive and guide it and constrain it, we can create a much more sustainable economy in a world that works for everyone. ■



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